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SARDAR PATEL UNIVERSITY

S.Y.B.C.A. 4th SEMESTER (NC) CBCS Examination April - 2023

US04CBCA25 Data Structure - II

Date: 06/04/2023 Thursday

Time: 02:00pm to 05:00pm **Maximum Marks: 70**

Q.1 Answer the following questions. **[10]**

1 A binary tree has at most ____________ child.
(a) One (b) Two (c) Three (d) None of these

2 __________ of a node is the immediate predecessor of a node.
(a) Height (b) Parent (c) Level (d) Branch

3 __________ is a set of disjoint trees.
(a) Forest (b) Path (c) Directed tree (d) Height

4 Single linked list is also known as ____________.
(a) One-way list (b) Two-way list (c) Three-way list (d) Four-way list

5 LLINK / LPTR is the pointer pointing to the ___________.
(a) successor node (b) head node (c) predecessor node (d) last node

6 The minimum number of fields with each node of double linked list is ____
(a) 1 (b) 2 (c) 3 (d) 4

7 ________ is the process to find the particular element from the array.
(a) Sorting (b) Inserting (c) Updating (d) Searching

8 Which design technique is used for Quick Sort...?
(a) Backtracking (b) Greedy (c) Dynamic Programming (d) Divide and Conqueror

9 ___________ file is also known as Random file.
(a) Direct (b) Sequential (c) Serial (d) None of these

10 __________ indicates how records are distributed over a no. of cylinders.
(a) Cylinder index (b) Prime entry (c) Track index (d) Master index

Q.2 Explain following in brief. (Attempt any Ten) **[20]**

1 Define Following Terms: Leaf , Path .

2 Draw the Binary Tree for (A-B)+C*(E/F)

3 List out Applications of Tree.

4 Explain linked list in brief.

5 Explain doubly linked list in brief.

6 List delete operations that can be performed on singly linked list.

7 Define sorting. List out the names of sorting techniques.

8 Discuss the application of searching.

9 Define: Internal Sort and External Sort.

10 Explain mid square hashing function.

11 Define Bucket Capacity.

12 Explain in brief index area.

[P.T.O.]

Q-3 (A) Write down an algorithm for inorder traversal of a binary tree. [05]

(B) Draw the binary tree for following expressions: [05]

Inorder : n1 n2 n3 n4 n5 n6 n7 n8 n9

Postorder : n1 n3 n5 n4 n2 n8 n7 n9 n6

OR

Q-3 (A) Explain the insertion of a node in a binary with an example [05]

(B) Write down an algorithm for postorder traversal of a binary tree. [05]

Q-4 (A) Write an algorithm to delete node from any position in singly linked list. [05]

(B) Write an algorithm to insert node at any position in singly linked list [05]

OR

Q-4 (A) Write an algorithm to insert node at front position in singly linked list. [05]

(B) List and explain types of linked lists with example. [05]

Q-5 (A) Write down the algorithm of Binary Search. [05]

(B) Write down the algorithm of Merge sort. [05]

OR

Q-5 (A) Differentiate between Sorting and Searching. [05]

(B) Write down the algorithm of bubble sort. [05]

Q-6 List and explain popular hashing function with example. [10]

OR

Q-6 Write a detail note on structure and processing of index sequential file supported by IBM [10]

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